

Jevi Waugh

0420 463 525 | schallum0712@gmail.com | linkedin.com/in/jevi-waugh | [Github: Jevi-Waugh](https://github.com/Jevi-Waugh)

[Website: jeviwaugh.com](https://jeviwaugh.com) | [ORCID Wandb.ai](https://orcid.org/Wandb.ai)

EDUCATION

The University of Queensland

Brisbane, QLD

Master of Data Science – Semester 1 GPA: 6.5/7.0

2026 – Present

The University of Queensland

Brisbane, QLD

Bachelor of Computer Science, Major in Machine Learning (Semester 2 GPA - 6.5 / 7.0)

2024 – 2025

PROFESSIONAL EXPERIENCE

AI Engineer

July 2026 – Present

Apex Fincap

Brisbane, QLD

- Contribute to the development of a fintech trading platform for mortgage brokers and finance professionals, working across application logic, data handling, and platform functionality.
- Support the integration of intelligent and automated platform features while contributing more broadly to software engineering and product development.
- Work under a non-disclosure agreement (NDA), limiting disclosure of specific system architecture and implementation details.

Machine Learning Research Assistant

Mar. 2026 – September 2026

The University of Queensland, School of EECS

Brisbane, QLD

- Derived **3 novel/new algorithms** such as **Kaleidoscope Fourier**, **Generalised Kaleidoscope Fourier** and **Fourier Unshuffle** for use in frequency-based autoencoders for image segmentation tasks.
- Developed and evaluated multiple autoencoder architectures to assess compatibility with novel frequency-domain algorithms, comparing model behaviour and performance across architectural variants.
- Contributing to research targeting manuscript submission in 2027, focusing on frequency-domain methods for computer vision.

Casual Academic/ Student assistant

Jul. 2025 – Present

The University of Queensland, School of EECS

Brisbane, QLD

- Deliver tutorials and practical classes, support students with programming and theoretical concepts, and assist with assessment marking and feedback.
- **July 2025 – Present:** CSSE1001 Introduction to Software Engineering
- **Semester 2, 2026 - Present:** COMP3710 Pattern Recognition and Analysis, DECO3801/7381 Design Computing Studio 3 - Build (Captone)
- **Semester 1, 2026:** COMP4702/7703 Machine Learning, COMP3400 Functional and Logic Programming, COMP2048 Theory of Computing

Summer Research Scholar

Jan. 2026 – Mar. 2026

The University of Queensland, School of EECS

Brisbane, QLD

- Completed a competitive program that led to an offer for a subsequent Research Assistant position.
- Designed and implemented a **PsychoNet-based decoder** for medical image segmentation and derived the **Fourier Shuffle** algorithm for frequency-spatial upsampling.
- Implemented and evaluated the architecture in **PyTorch**, benchmarking against a state-of-the-art U-Net model and achieving a **1.77% improvement in validation performance** with a **13.8% reduction in GPU memory usage under specific parameters**.
- Awarded a \$3,000 SRS grant to conduct research in machine learning and medical image segmentation.
- Developed custom training and evaluation pipelines for GPU compute nodes under the supervision of [Shekhar Chandra](#) and [Wendi Ma](#).

VOLUNTEERING EXPERIENCE / EXTRA CURRICULAR

Industry Officer

Nov. 2025 – Present

UQ Computing Society

Brisbane, QLD

- Managed UQCS industry partnerships, including sponsor outreach, pitching, negotiation, relationship management, and partnership development.
- Secured over **\$35,000 in sponsorship funding** for student initiatives and industry engagement programs.
- Increased sponsorship funding by **43% in 2026** while expanding the organisation's sponsor base by **30%**.
- Onboarded new industry partners including **Atlassian**, **Citadel**, **Cybermetrix**, **getcracked.io**, **Quantum Australia** and **Tanda**. My industry report will be available through this link by end of October.

PROJECTS

Vision Transformer Attention Analysis | *Python, PyTorch, Hugging Face*

Jul. – Dec. 2025

- Investigated internal attention dynamics of Vision Transformer architectures using pretrained [DeiT models](#).
- Built a pipeline for extracting and analysing attention distributions across transformer layers.
- Conducted experiments modifying attention formulations to study model behaviour in collaboration with [Dr. Kasra Khosoussi](#).

Fine-tuning FLAN-T5 for Biomedical Lay Summarisation | *PyTorch, Transformers, NLP*

- Fine-tuned **FLAN-T5** encoder-decoder models on the **BioLaySumm** dataset using mixed-precision training.
- Evaluated generated summaries using **ROUGE-1, ROUGE-2, ROUGE-L, and ROUGE-Lsum**.
- Achieved **greater than 70% ROUGE performance** across evaluation metrics on held-out biomedical summarisation tasks.
- Project available through this link

TraceForest: AI-Generated Code Detection | *Python, Java, Random Forest, SHAP*

- Developed an explainable Random Forest classifier achieving **94% accuracy** for AI-generated code detection using stylometric AST features and **SHAP-based explainability**. This was for my capstone course.
- Built Python and Java adapters returning classification probabilities for automated pull-request review workflows.
- Extended [CodeGPTSensor](#) through encoder fine-tuning with stylometric features, achieving **greater than 95% accuracy**.
- Released the system as a **VS Code extension with 3,000+ downloads**. [VS Code Marketplace](#)

TECHNICAL SKILLS

Languages: Python, C/C++, Java, Haskell, SQL, Mojo (familier)

Machine Learning / Data: PyTorch, scikit-learn, Einops, Pandas, NumPy, Matplotlib, OpenCV

Data Engineering: Hadoop, PySpark

Developer Tools: Git, GitHub, Linux, Jira